

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without permission of BIS or used as an Indian Standard)

भारतीय मानक मसौदा
मेयो पैटर्न कैंची, विच्छेदन, सीधे और फ्लैट पर घुमावदार —
विशिष्टि

(IS 9146 का पहला पुनरीक्षण)

Draft Indian Standard

**Mayo Pattern Scissors, Dissecting, Straight and
Curved on Flat — Specification**

(First Revision of IS 9146)

ICS 11.040.30

Surgical Instruments Sectional Committee, MHD 01 Last date for comments: 04 October 2026

FOREWORD

(Formal clauses will be added later)

This Indian Standard was first published in 1979 as ‘Specification for Scissors, Dissecting, Straight and Curved on Flat Mayo’s Pattern’. This revision aligns the cross references to the latest standards, incorporates revised designations for steel and revised certification clause.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (second revision)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This Indian Standard covers dimensional and other requirements for scissors, dissecting, straight and curved on flat, Mayo's pattern used in general surgery.

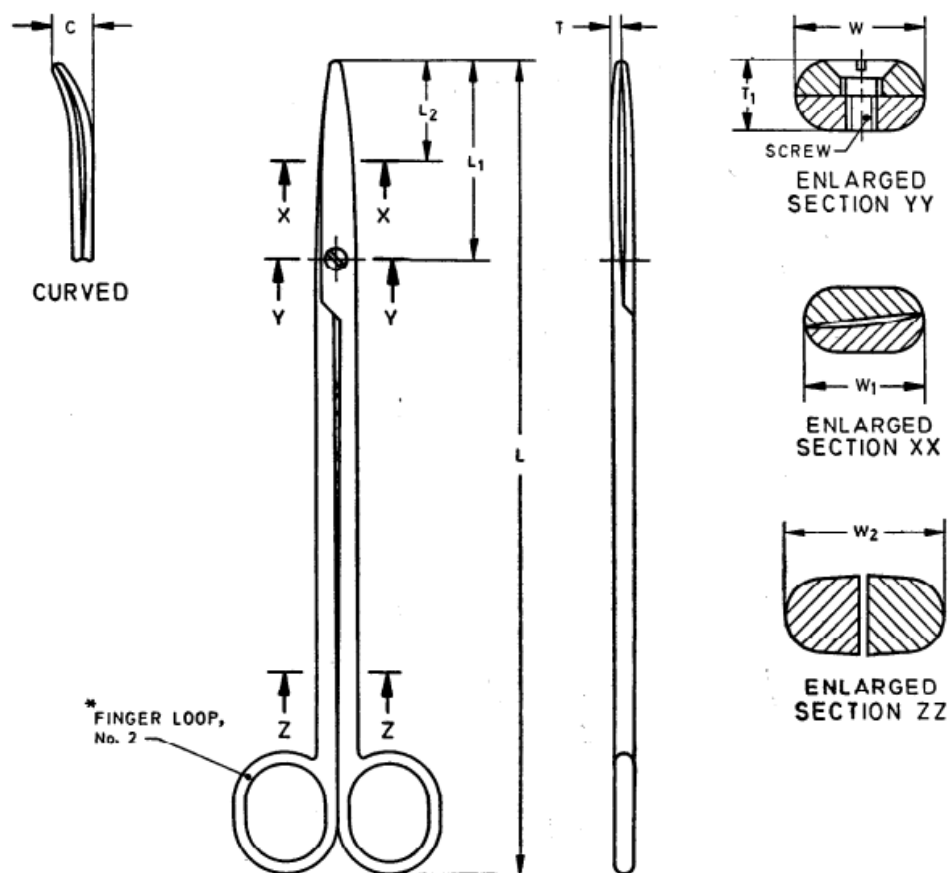
2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

IS No.	Title
IS 6603 : 2024	Stainless Steel Semi-Finished Products, Bars, Wire Rods and Bright Bars - Specification (<i>second revision</i>)
IS 7531 : 1990	Surgical Instruments - Corrosion resistance of Stainless Steel Surgical Instruments - Methods of Tests (<i>first revision</i>)

3 SHAPE AND DIMENSIONS

Shape and dimensions shall be as shown in Fig. 1.



Size	L $\pm$ 5	L $_1\pm$ 2	L $_2\pm$ 1	W $\pm$ 0.5	W $_1\pm$ 0.5	W $_2\pm$ 0.5	T $\pm$ 0.2	T $_1\pm$ 0.2	C $\pm$ 0.5	Screw
1	140	45	25	10.5	8.0	10.5	1.6	5.5	11.5	M2.5
2	165	65	32	11.0	8.7	11.2	2.4	5.5	15.5	M2.5
3	190	70	32	12.5	9.5	12.7	2.4	6.5	17.5	M3
4	215	80	38	13.5	10.5	14.3	2.4	7.0	19.0	M3

All dimensions in millimetres.

FIG.1 MAYO PATTERN SCISSORS, DISSECTING, STRAIGHT AND CURVED ON FLAT

4 MATERIALS

Scissors shall be made of stainless steel conforming to Designation X30Cr13 of IS 6603 or to the following composition:

<i>Element</i>	<i>Percent</i>
Carbon	0.35 to 0.45
Silicon	0.10 to 0.70
Manganese	0.30 to 0.70
Nickel	1.00 <i>Max</i>
Chromium	12.00 to 14.00

5 WORKMANSHIP AND FINISH

5.1 Scissors shall be symmetrical and well formed.

5.2 The blades along their length and the tips shall coincide and shall not over-ride, when the scissors are fully closed.

5.3 The blades shall open and close without stiffness and shall have a slight cross-over action to give a continuous pinch, and shall have no play at the joint.

5.4 Scissors shall be free from cracks, seams, burrs, flaws and other defects. The surfaces of the scissors shall be finished smooth, and polished bright passivated. The edges shall be even and rounded except for the cutting edge which shall be sharp. The cutting edge shall not have any feather, and shall be made hollow ground.

6 HARDNESS

Scissors shall be evenly hardened and tempered to give a hardness of 550 to 650 *HV*.

7 TESTS

7.1 Performance Test

The scissors shall cut wet tissue paper, teased - out cotton wool, fine hair and frayed lint accurately and cleanly from pinch to tip.

7.2 Hardness Test

When tested on a hardness testing machine, the scissors shall have a Vicker's diamond pyramid hardness number as specified in 5.

7.3 Flexibility Test

The flexibility shall be tested as given in 7.3.1 and 7.3.2.

7.3.1 The finger ring of one arm of the scissors shall be fixed in a vice, with the shank and the blade projecting above the vice. By the application of force at a point 75 mm above the vice, the shank shall be deflected at right angles to the plane of the finger loop to a distance of 10 mm as measured at the tip of the blade. On release of the force, no permanent set shall be observed. The test shall be carried out on other arm also.

7.3.2 The test shall be repeated on the same arm with the arm fixed in the vice near the joint so that 75 mm of the shank (as measured from the upper surface of the vice jaws to the upper pole of the finger loop) protrudes above the vice. The deflecting force shall be applied at the upper pole of the finger loop and shall act in a plane at right angles to the plane of the finger loop. The shank shall be deflected by 5 mm as measured at the upper pole of the finger loop. On release of force, no permanent set shall be observed. The test shall be carried out on the other arm also.

7.4 Corrosion Resistance Test

Test the scissors in accordance with IS 7531. It shall show no sign of corrosion after the test.

8 MARKING

8.1 The scissors shall be legibly and indelibly marked with the manufacturer's name, initials or recognized trade-mark and the words 'Stainless Steel'. The marking shall preferably be done by electrical etching and not by stamping.

8.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

9 PACKING

Each scissors may be wrapped in moisture-proof polyethylene bag and packed in carton in a manner that coming into contact with one another is avoided. The scissors may also be packed as agreed to between the purchaser and the manufacturer.